

Coding & Solution

Date	29 June 2026
Team ID	
Project Name	Rising Waters: AI-Powered Flood Risk Prediction System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sasivardhan3011/Rising-Waters
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib
Key Features Implemented	- Flood risk prediction (Low, Moderate, High) using hydrological parameters. - Input of rainfall, river level, land use, soil type, and other relevant data. - Machine learning-based flood risk prediction with probability score. - Risk probability analysis and flood risk level classification. - Interactive dashboards with charts and graphs for data visualization. - Responsive web interface with Flask API integration for real-time predictions.
Pending / Incomplete Features	- Real-time weather API integration - Early warning alert system (Email / SMS) - Geospatial flood mapping - Mobile application support
Setup / Run Instructions	Install the required dependencies, run the Flask backend (python app.py), launch the React application (or open the deployed Vercel URL), and enter hydrological data to use the flood risk prediction and analysis features.

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Rising Waters project is designed to predict flood risk levels by analyzing hydrological parameters such as rainfall, river level, land use, and soil type. The application follows a simple and organized structure, making it easy to understand, maintain, and improve. Machine learning is used to generate accurate flood risk predictions, while Flask connects the prediction model with a user-friendly web interface. The project can be extended in the future by adding real-time weather integration, early warning alerts, geospatial flood mapping, and mobile application support.